IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

SEMICONDUCTOR DEVICES

Thursday, 8 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Q1 is compulsory. Answer Q1 and any two of questions 2-4

Q1 carries 40% of the marks. Questions 2 to 4 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Z, Durrani
Second Marker(s) :

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Constants

Permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
Permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
Intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$ K
Dielectric constant of Si:	$\varepsilon_{Si} = 11$
Dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
Thermal voltage:	$kT/e = 0.026$ V at $T = 300$ K
Charge of an electron:	$e = 1.6 \times 10^{-19}$ C
Planck's constant:	$h = 6.63 \times 10^{-34}$ Js
Boltzmann constant:	$k = 1.38 \times 10^{-23}$ J/K
Band gap of Si:	$E_G = 1.12$ eV at $T = 300$ K
Electron affinity of Si:	$\chi = 4.05$ eV at $T = 300$ K
Effective density of states of Si:	$N_C = 3.2 \times 10^{-19}$ cm^{-3} at $T = 300$ K $N_V = 1.8 \times 10^{-19}$ cm^{-3} at $T = 300$ K

Formulae

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$ Schrödinger's equation in one dimension

$$f(E)=\frac{1}{1+\exp\left(\dfrac{E-E_f}{kT}\right)}$$ Fermi distribution

$$m_e^*=\frac{\hbar^2}{d^2E(k)/dk^2}$$ Effective mass

$$n=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_e^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_c-E_F}{kT}\right)}=N_ce^{\left(-\frac{E_c-E_F}{kT}\right)}$$ Concentration of electrons

$$p=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_h^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_F-E_v}{kT}\right)}=N_ve^{\left(-\frac{E_F-E_v}{kT}\right)}$$ Concentration of holes

$$\left.\begin{aligned}J_n(x)&=e\mu_n n(x)E(x)+eD_n\frac{dn(x)}{dx}\\J_p(x)&=e\mu_p p(x)E(x)-eD_p\frac{dp(x)}{dx}\end{aligned}\right\}$$ Drift and diffusion current densities in a semiconductor

$$I_{DS}=\frac{\mu C_{ox}W}{L}\left((V_{GS}-V_{th})V_{DS}-\frac{V_{DS}^2}{2}\right)$$ Current in a MOSFET

$$\left.\begin{aligned}J_n&=\frac{eD_n n_{p_0}}{L_n}\left(e^{\frac{eV}{kT}}-1\right)\\J_p&=\frac{eD_p p_{n_0}}{L_p}\left(e^{\frac{eV}{kT}}-1\right)\end{aligned}\right\}$$ Diffusion current densities in a pn-junction

$$V_0=\frac{kT}{e}\ln\left(\frac{N_AN_D}{n_i^2}\right)$$ Built-in voltage

$$c=c_0\exp\left(\frac{eV}{kT}\right)\text{ with }\begin{cases}c=p_n\text{ or }n_p\\c_0\text{ bulk minority carrier concentration}\end{cases}$$ Minority carrier injection under bias V

$$D=\frac{kT}{e}\mu$$ Einstein relation

$$w_n=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_A}{N_AN_D+N_D^2}(V_0-V)}$$

$$w_p=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_D}{N_AN_D+N_A^2}(V_0-V)}$$ Depletion width in the n and p-type region, under bias V

$$\delta c = \Delta c \exp\left(\frac{-x}{L}\right) \text{ with } \begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$$

Excess carrier concentration as a function of distance when recombination occurs – long layer approximation.

$$L = \sqrt{D\tau}$$

Diffusion length

$$C_{diff} = \frac{e}{kT} I\tau$$

Diffusion capacitance

$$i(t) = \frac{Q(t)}{\tau} + \frac{dQ(t)}{dt}$$

Time variation of current and charge

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon}$$

Poisson equation in 1 dimension

$$n = N_C \, exp\left(\frac{E_F - E_C}{kT}\right)$$

$$p = N_V \, exp\left(\frac{E_V - E_F}{kT}\right)$$

Carrier concentrations

1.

a) (i) Explain briefly how minority carriers move, under forward bias, from the junction region to the contacts in a diode.

[4]

(ii) Define the long- and short-diode approximations, using the relevant semiconductor parameters for these configurations.

[4]

b) (i) Write down the general expression for charge neutrality in a doped semiconductor.

[4]

(ii) Hence, for the case of a *n*-type, compensated semiconductor with doner concentration N_D, by using the Law of Mass Action, show that the electron concentration *n* may be approximated as:

$$n = \frac{N_D}{2} + \frac{N_D}{2}\sqrt{1 + \frac{4n_i^2}{N_D^2}}$$

[6]

c) (i) A *pn* diode is *p* doped at 10^{16} cm^{-3} and *n* doped at 10^{15} cm^{-3}. Under the long diode approximation, draw diagrams showing the excess electron and hole minority carrier densities along the length of the diode, under forward and reverse bias.

[4]

(ii) For forward and reverse bias magnitudes of 0.5 V on this diode, calculate the excess minority carrier concentrations at the depletion region edges.

[6]

d) Sketch the band diagram (E_c, E_v, E_F, E_G), from the emitter to the collector, for a p^+np bipolar junction transistor (BJT) with:

i) The BJT at equilibrium

[4]

ii) The BJT in cut-off.

[4]

iii) The BJT in forward active mode.

[4]

2.

a) Consider a *n*-type 1D semiconductor wire, with cross-sectional area A and length along x-axis. An excess minority carrier concentration $\delta p(x)$ exists along the wire. By considering the hole diffusion current density $J_p(x)$ along the wire, derive the following expression:

$$\frac{d\delta p}{dt} = D_p \frac{d^2\delta p}{dx^2}$$

State any assumptions you make.

[10]

b) A sample of intrinsic semiconductor is doped inhomogeneously with donors along the x direction. The distribution of the doping atom concentration N_D is given by the function:

$$N_D(x) = N_0 \exp(-ax)$$

(i) Derive an expression for the built-in electric field with zero external voltage applied, and $N_D >> n_i$. What is the magnitude of this field for $a = 2\ \mu m^{-1}$?

[6]

(ii) Sketch the energy band diagram for this semiconductor

[3]

(iii) Briefly discuss your interpretation of this diagram.

[3]

c) A p^+np Si bipolar transistor has a uniform cross-sectional area for current flow of 2×10^{-4} cm^2 and a base width of 2 μm. The emitter doping is 5×10^{18} cm^{-3} and the base doping 10^{16} cm^{-3}. The hole and electron mobilities are μ_p = 400 cm^2/Vs and μ_n = 1000 cm^2/Vs, and temperature T =300 K. Find an expression for, and hence calculate I_C with V_{EB} = 0.7 V.

[8]

3.

a) What is the main recombination process for carriers in silicon semiconductors under low electric fields and why are other recombination processes less significant?

[6]

b) The time dependent excess hole charge $Q_p(t)$ in a p^+n diode is given by:

$$\frac{dQ_p(t)}{dt} = i_p - \frac{Q_p(t)}{\tau_p}$$

Here, τ_p is the hole recombination time and i_p the hole current.

The diode, with a resistive load, is to be switched 'off' by switching a forward bias to either (A) zero bias, or (B), reverse bias. Using the given expression for $Q_p(t)$:

(i) Find expressions for and sketch the time variation of the excess hole current for both conditions (A) and (B).

[12]

(ii) Show that at time T, when the excess charge falls to zero for condition (B), half the initial charge still remains for condition (A).

[6]

c) Briefly discuss any additional causes of time delays for a real, non-ideal diode.

[6]

4.

An *n*-channel Si MOSFET is shown in Figure 4.1. The device has gate length L, gate width W, gate oxide thickness t_{ox} and gate capacitance per unit area C_{ox}. The doping concentration in the source and drain regions is N_D (per unit volume). The bulk doping concentration is N_A (per unit volume). The drain-source and gate-source voltages are V_{ds} and V_{gs} respectively, the MOSFET threshold voltage in V_{th}, and the source is grounded.

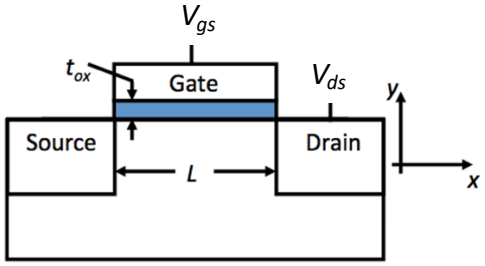


Figure 4.1

a) (i) Find an expression for the inversion layer charge per unit area, Q_n, as a function of the voltage drop $V_x(x)$ along the channel.

[6]

(ii) Provide a justification for your expression, including any approximations you have made.

[4]

b) Using your expression in (a), write down expressions for Q_n and the channel depth $a(x)$ at the source and drain ends of the channel region at pinch-off.

[8]

c) Hence, by considering the drift current in the channel, derive the following expression for the channel current I:

$$I = C_{ox}\mu_e \frac{W}{L}\left((V_{gs} - V_{th})V_{ds} - \frac{V_{ds}^2}{2}\right)$$

[12]

[END]